

Preference-Aware Coefficient Correction for Rewarded-Soups-Style Model Merging



Motivation and Goal

$$\theta_i \rightarrow \delta_i = \theta_i - \theta_0 \rightarrow R \rightarrow \lambda^* = f(p, R)$$

Motivation:

User preferences p are defined over objectives, but model merging happens in task-vector space.

Goal:

Use task-vector geometry R to compute preference-aware merge coefficients λ^* .

Architecture Overview + Method

- $\delta \in \mathbb{R}^P$

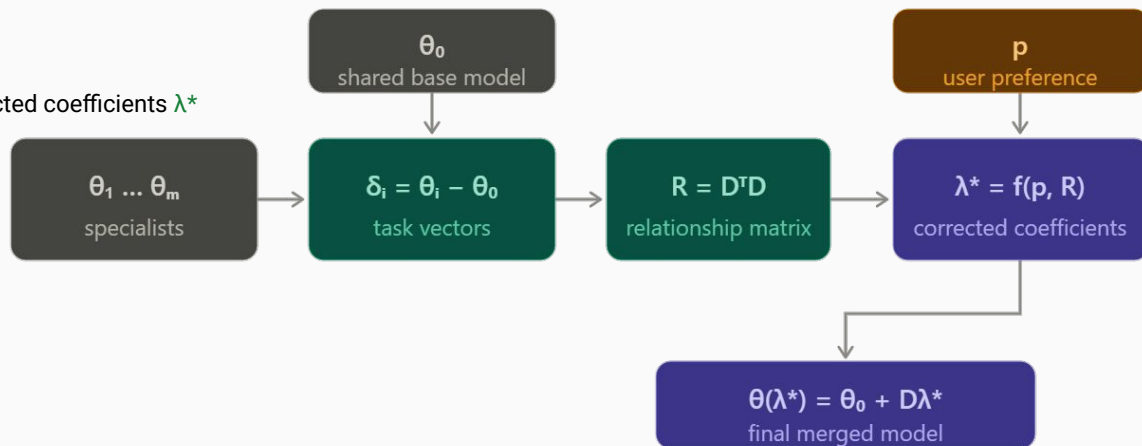
$$D = [\delta_1, \dots, \delta_m]$$

- $D \in \mathbb{R}^{(m \times P)}$
- Base model: TinyLlama-1.1B
- $R \in \mathbb{R}^{(m \times m)}$
- Five HelpSteer2 LoRA specialists
- Compute relationship matrix:
- Compute corrected coefficients:
- $\lambda \in \mathbb{R}^m$
- Merge adapters and evaluate with ArmoRM

$$R = D^\top D$$

$$\lambda^* = f(p, R)$$

1. Train attribute adapters
2. Extract task vectors
3. Compute relationship matrix R
4. Given preference vector p , solve for corrected coefficients λ^*
5. Merge adapters
6. Evaluate merged model



Difficulties

- Run proxy validation to check assumption
 - Takes ~10 hours
 - First run had cache contamination
- Data is highly correlated:
 - For helpfulness: Average complexity rating 3.96

```
helpfulness boundary counts (entry=count(correctness, coherence, complexity, verbosity exactly cap)):  
target\other_cap <=0 <=1 <=2 <=3 <=4  
rating=0 12(12,12,12,12) 69(6,51,39,25) 338(0,201,99,151) 815(1,391,36,152) 1616(2,741,6,97)  
rating=1 0(0,0,0,0) 11(10,10,9,8) 265(34,209,122,180) 936(15,559,50,226) 1952(52,902,5,170)  
rating=2 0(0,0,0,0) 0(0,0,0,0) 132(118,132,78,105) 1292(295,1058,83,263) 2445(82,1042,5,114)  
rating=3 0(0,0,0,0) 0(0,0,0,0) 6(5,6,4,4) 1621(1556,1583,195,315) 5877(1249,3594,23,100)  
rating=4 0(0,0,0,0) 0(0,0,0,0) 1(0,1,1,1) 12(9,10,1,2) 8434(8306,8266,66,84)
```

Adapter training setup

SFT Adapter Setup

- 5 LoRA adapters, one per HelpSteer2 axis
- Response-only SFT with prompt masking
- Same data size and hyperparameters across axes
- Best checkpoint selected by validation loss

PPO Setup

- Starts from shared `theta_SFT` initialization
- 5 PPO-LoRA adapters, one per ArmoRM head
- ArmoRM used as reward model, frozen during training
- Same query set, steps, and seeds across axes
- Used only as circular upper-bound experiment, not proxy validation

Adapter difficulties

- Did not work so far
- Debugging for 3 days

New Methods

$$\arg \max_{\lambda \in \Delta_m} p^\top R \lambda - \rho (\lambda - p)^\top R (\lambda - p), \quad \rho > 0$$

$$\arg \max_{\lambda \in \Delta_m} \min_i \Delta_i(\lambda) \quad \text{s.t.} \quad (\lambda - p)^\top R (\lambda - p) \leq c^2 \max(p^\top R p, \varepsilon)$$

Question

How is Naive and Oracle calculated?

Beaver					Assistant					Summary				
λ	Naive	Oracle	Gain	Gain%	λ	Naive	Oracle	Gain	Gain%	λ	Naive	Oracle	Gain	Gain%
[0.0, 1.0]	0.7924	0.8045	+0.0122	1.5%	[0.0, 0.0, 1.0]	0.8694	0.9110	+0.0417	4.8%	[0.0, 0.0, 1.0]	0.6502	0.7739	+0.1237	19.0%
[0.1, 0.9]	0.7405	0.7632	+0.0227	3.1%	[0.0, 0.2, 0.8]	0.7405	0.8143	+0.0738	10.0%	[0.0, 0.2, 0.8]	0.5965	0.7213	+0.1247	20.9%
[0.2, 0.8]	0.6851	0.7266	+0.0415	6.1%	[0.0, 0.4, 0.6]	0.6270	0.7301	+0.1031	16.4%	[0.0, 0.4, 0.6]	0.5743	0.6944	+0.1201	20.9%
[0.3, 0.7]	0.6298	0.6941	+0.0643	10.2%	[0.0, 0.6, 0.4]	0.5527	0.6596	+0.1069	19.3%	[0.0, 0.6, 0.4]	0.5955	0.7027	+0.1072	18.0%
[0.4, 0.6]	0.5901	0.6675	+0.0774	13.1%	[0.0, 0.8, 0.2]	0.5297	0.6188	+0.0891	16.8%	[0.0, 0.8, 0.2]	0.6586	0.7413	+0.0827	12.6%
[0.5, 0.5]	0.5823	0.6545	+0.0721	12.4%	[0.0, 1.0, 0.0]	0.5411	0.6082	+0.0671	12.4%	[0.0, 1.0, 0.0]	0.7357	0.8025	+0.0668	9.1%
[0.6, 0.4]	0.6055	0.6571	+0.0516	8.5%	[0.2, 0.0, 0.8]	0.8355	0.8839	+0.0485	5.8%	[0.2, 0.0, 0.8]	0.6204	0.7276	+0.1072	17.3%
[0.7, 0.3]	0.6253	0.6702	+0.0448	7.2%	[0.2, 0.2, 0.6]	0.7086	0.7758	+0.0671	9.5%	[0.2, 0.2, 0.6]	0.5654	0.6748	+0.1095	19.4%
[0.8, 0.2]	0.6441	0.6875	+0.0434	6.7%	[0.2, 0.4, 0.4]	0.6046	0.6888	+0.0842	13.9%	[0.2, 0.4, 0.4]	0.5505	0.6552	+0.1048	19.0%
[0.9, 0.1]	0.6672	0.7068	+0.0396	5.9%	[0.2, 0.6, 0.2]	0.5495	0.6273	+0.0778	14.2%	[0.2, 0.6, 0.2]	0.5829	0.6842	+0.1013	17.4%
[1.0, 0.0]	0.6840	0.7276	+0.0435	6.4%	[0.2, 0.8, 0.0]	0.5449	0.6105	+0.0656	12.0%	[0.2, 0.8, 0.0]	0.6521	0.7510	+0.0989	15.2%
					[0.4, 0.0, 0.6]	0.8065	0.8656	+0.0590	7.3%	[0.4, 0.0, 0.6]	0.5758	0.6864	+0.1105	19.2%
					[0.4, 0.2, 0.4]	0.6860	0.7498	+0.0638	9.3%	[0.4, 0.2, 0.4]	0.5431	0.6405	+0.0974	17.9%
					[0.4, 0.4, 0.2]	0.5974	0.6567	+0.0593	9.9%	[0.4, 0.4, 0.2]	0.5479	0.6656	+0.1176	21.5%
					[0.4, 0.6, 0.0]	0.5621	0.6205	+0.0584	10.4%	[0.4, 0.6, 0.0]	0.6001	0.7486	+0.1485	24.7%
					[0.6, 0.0, 0.4]	0.7812	0.8512	+0.0700	9.0%	[0.6, 0.0, 0.4]	0.5676	0.6658	+0.0982	17.3%
					[0.6, 0.2, 0.2]	0.6759	0.7344	+0.0585	8.7%	[0.6, 0.2, 0.2]	0.5948	0.6930	+0.0982	16.5%
					[0.6, 0.4, 0.0]	0.6045	0.6570	+0.0526	8.7%	[0.6, 0.4, 0.0]	0.6583	0.7822	+0.1238	18.8%
					[0.8, 0.0, 0.2]	0.7729	0.8407	+0.0678	8.8%	[0.8, 0.0, 0.2]	0.6733	0.7352	+0.0619	9.2%
					[0.8, 0.2, 0.0]	0.6805	0.7335	+0.0530	7.8%	[0.8, 0.2, 0.0]	0.7610	0.8223	+0.0613	8.1%
					[1.0, 0.0, 0.0]	0.7769	0.8376	+0.0607	7.8%	[1.0, 0.0, 0.0]	0.8294	0.8656	+0.0362	4.4%

Next Steps

- Train Adapters similar as in Rewarded Soups
- Test out Methods on new R